PGCP-AI

PRACTICAL MACHINE LEARNING

Day 4: Regression

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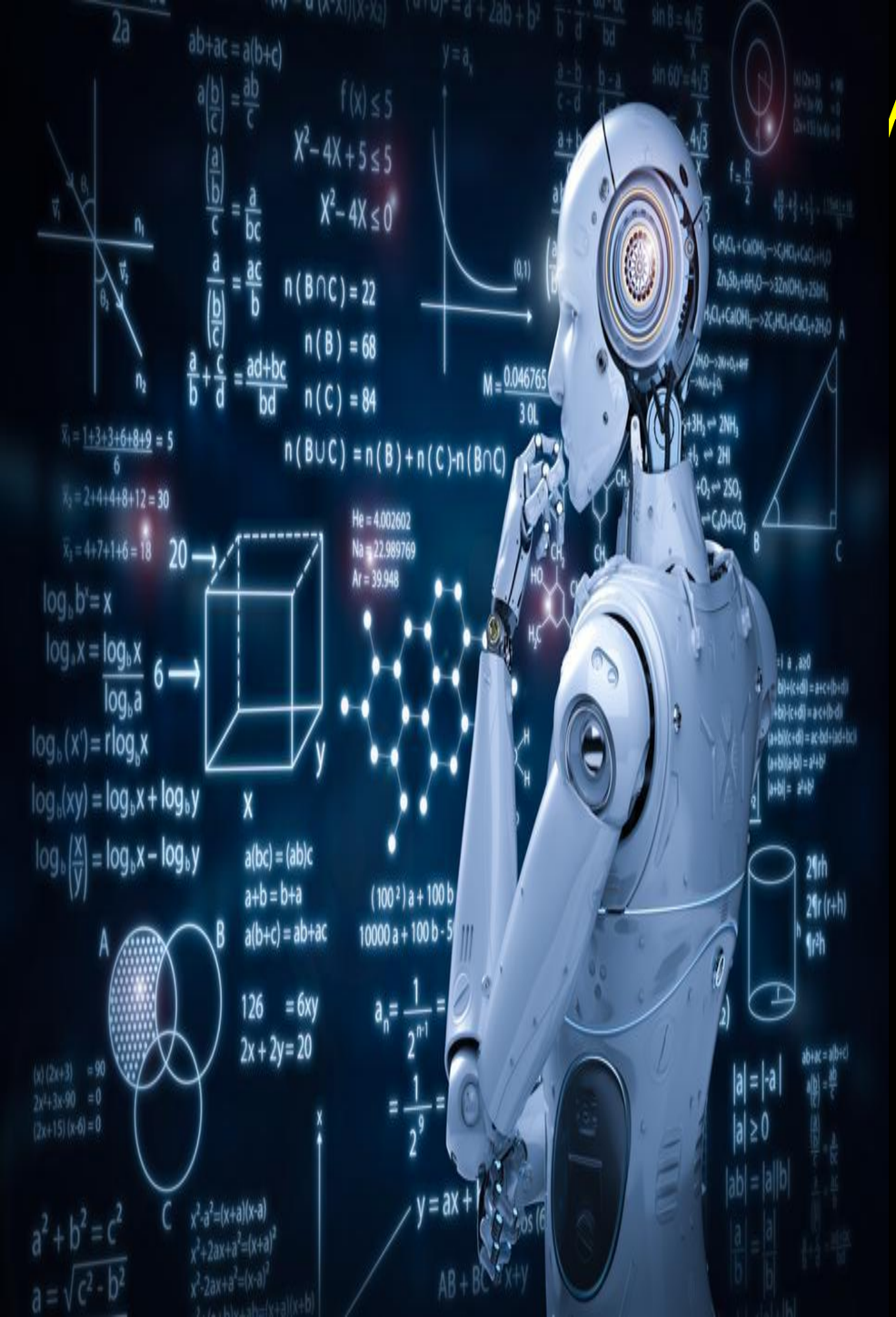
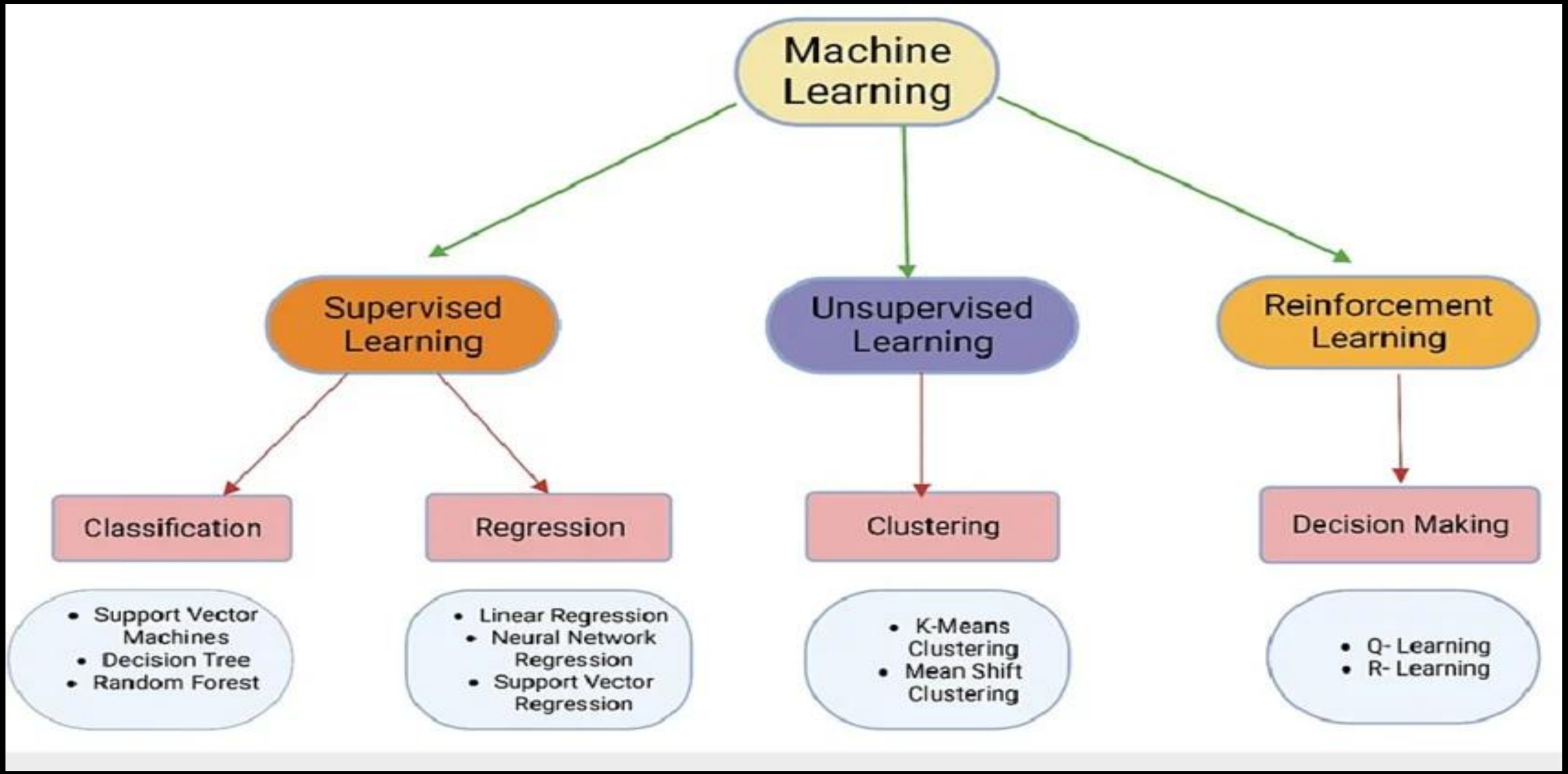


Agenda

- ML Algorithms
- Regression
- Types of Regression

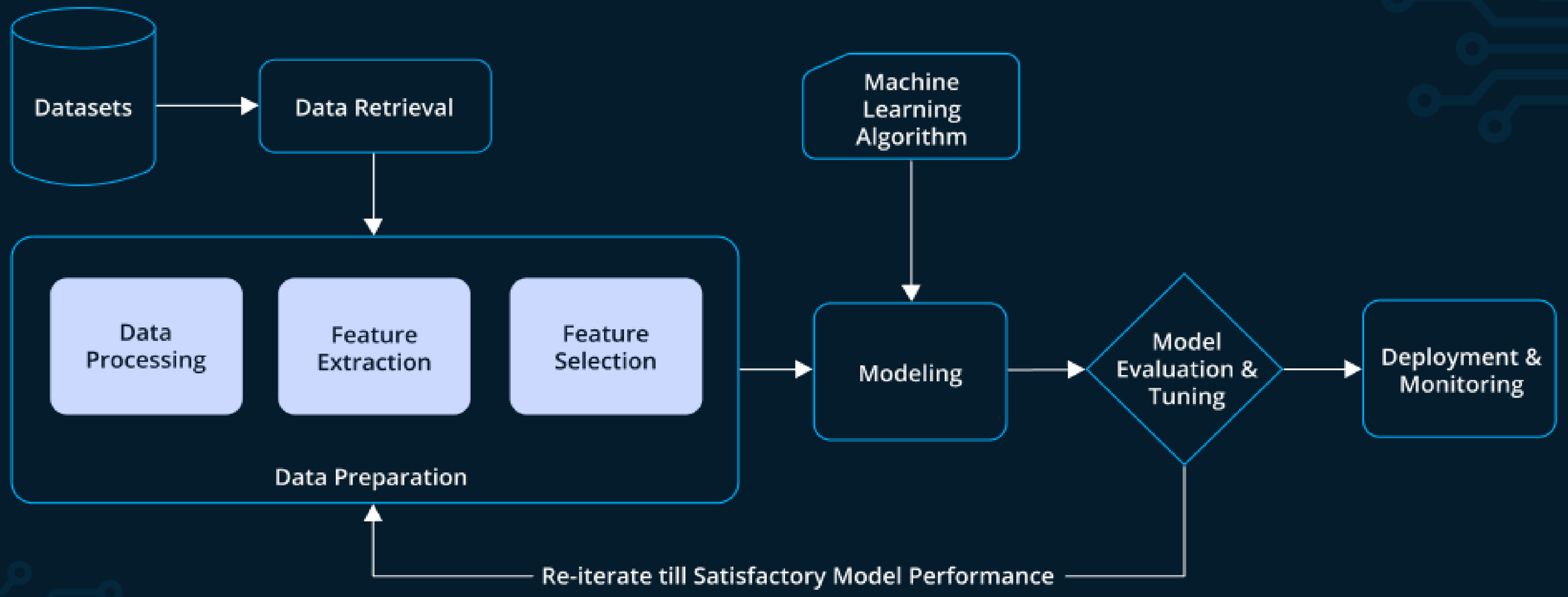
Learning Algorithms



Supervised Learning

- Supervised learning is a process of providing input data as well as correct output data to the machine learning model.
- The aim of a supervised learning algorithm is to **find a mapping function to map the input variable(x) with the output variable(y)**.
- In the real-world, supervised learning can be used for **Risk Assessment, Image classification, Fraud Detection, spam filtering**, etc.

Supervised Machine Learning



Leewayhertz

Supervised learning

- **Supervised learning** is a type of machine learning where a model is trained using **labeled data**—that is, each input comes with a known correct output. The goal is to learn a mapping from inputs to outputs so the model can make predictions on new, unseen data.
- **Core idea**
- You provide the model with examples:
- Input (features) → Output (label)
- The model learns the relationship between them and generalizes it.

How it works (basic steps)

- Collect labeled data
- Choose a model (e.g., linear regression, decision tree)
- Train the model on data
- Evaluate performance (accuracy, error, etc.)
- Use it to predict new data

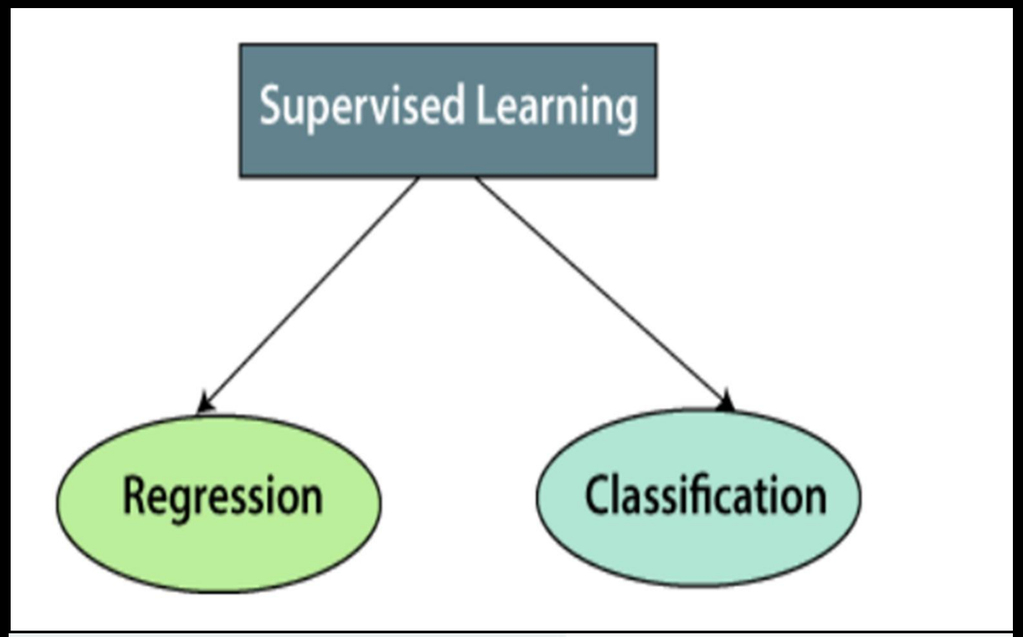
Advantages & Disadvantages

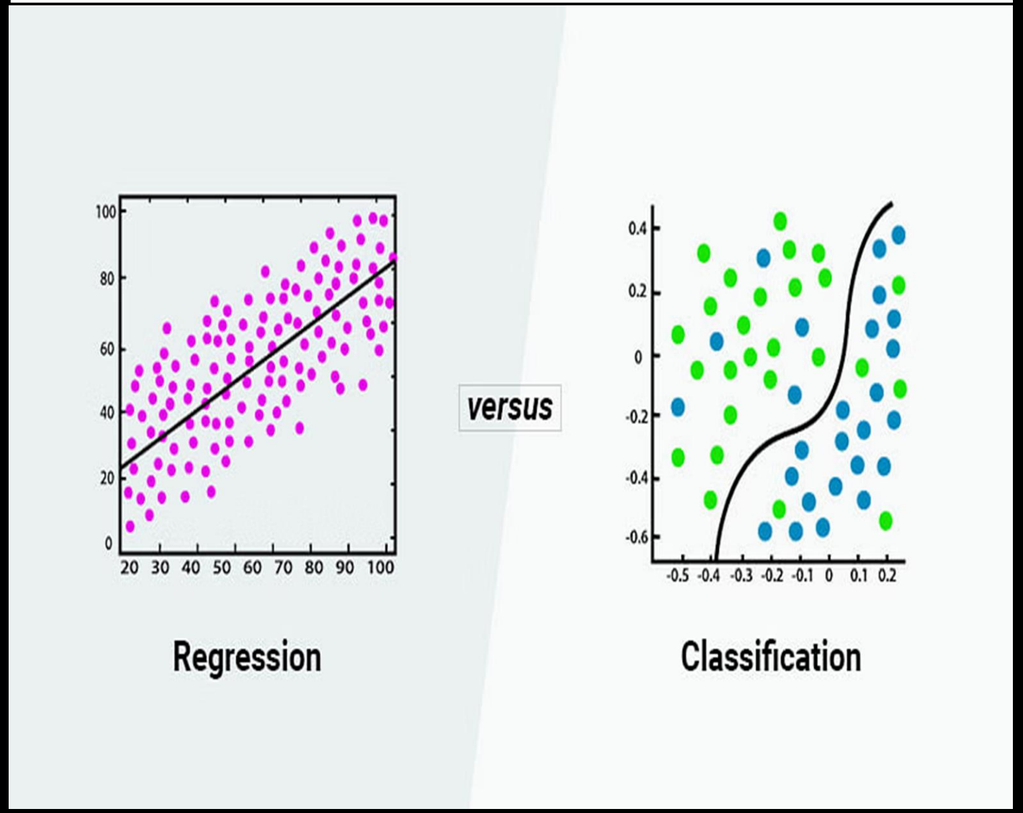
- **Key advantages**
- High accuracy when good labeled data is available
- Clear evaluation since correct answers are known

- **Limitations**
- Requires large amounts of labeled data
- Can overfit if not properly managed
- Performance depends on data quality

Types of supervised learning

- **Classification**
  - Output is a category (e.g., spam or not spam)
  - Examples: email filtering, disease detection
- **Regression**
  - Output is a continuous value (e.g., temperature, price)
  - Examples: stock price prediction, sales forecasting





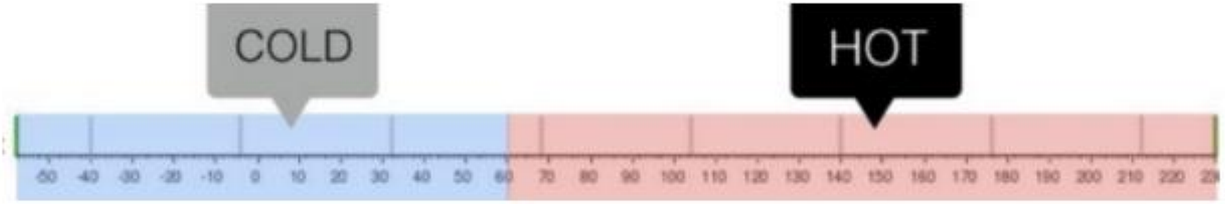
Regression

What will be the temperature tomorrow?

Fahrenheit

Classification

Will it be hot or cold tomorrow?



Fahrenheit

Regression

- Regression is used to study the relationship between two variables.
- We can use simple regression if both the dependent variable (DV) and the independent variable (IV) are numerical.
- If the DV is numerical but the IV is categorical, it is best to use ANOVA.

1. **Simple linear regression**

   Assume that there is only one independent variable x. If the relation between x and y is modeled by the relation

   $$y = a + bx$$

   then we have a simple linear regression.

2. **Multiple regression**

   Let there be more than one independent variable, say x_1, x_2, ..., x_n, and let the relation between y and the independent variables be modeled as

   $$y = \alpha_0 + \alpha_1 x_1 + \cdots + \alpha_n x_n$$

   then it is case of multiple linear regression or multiple regression.

3. **Polynomial regression**

   Let there be only one variable x and let the relation between x y be modeled as

   $$y = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

   for some positive integer $n > 1$, then we have a polynomial regression.

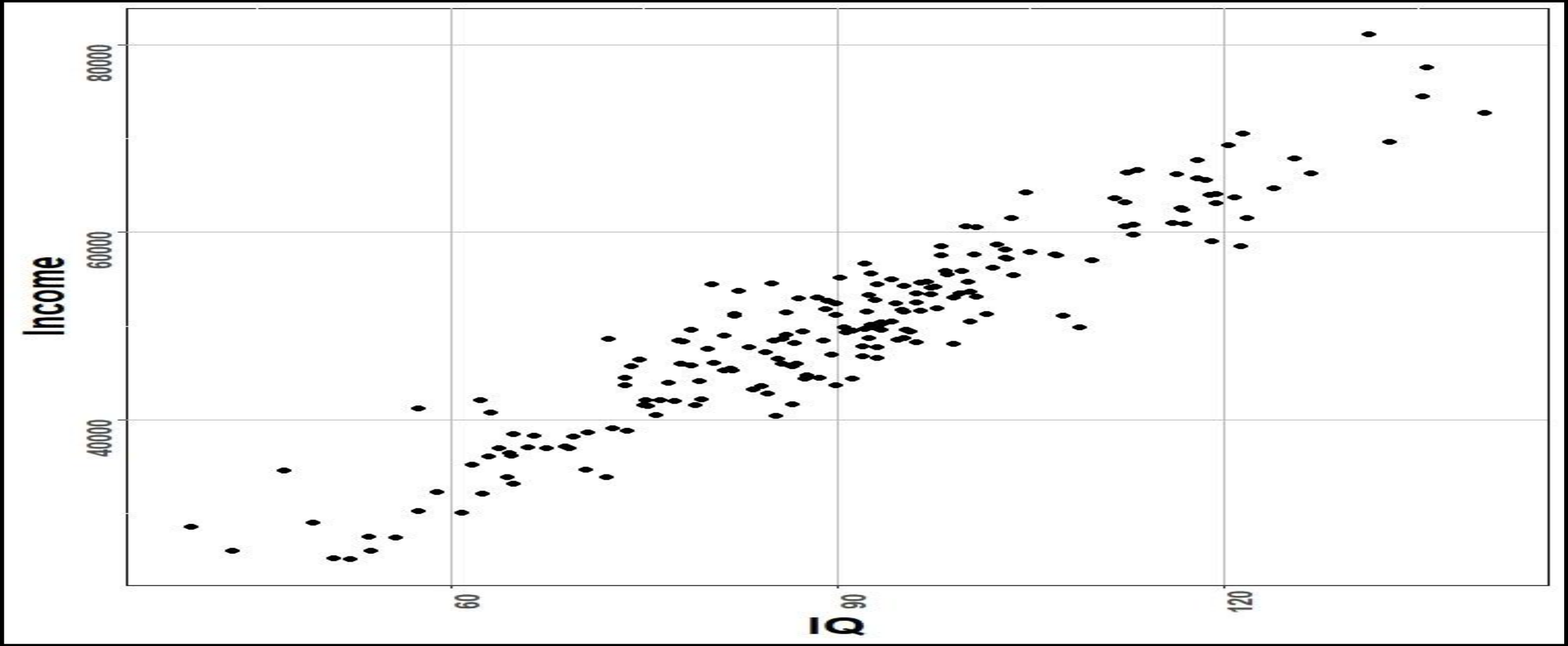
4. **Logistic regression**

   Logistic regression is used when the dependent variable is binary (0/1, True/False, Yes/No) in nature. Even though the output is a binary variable, what is being sought is a probability function which may take any value from 0 to 1.

Simple Linear Regression

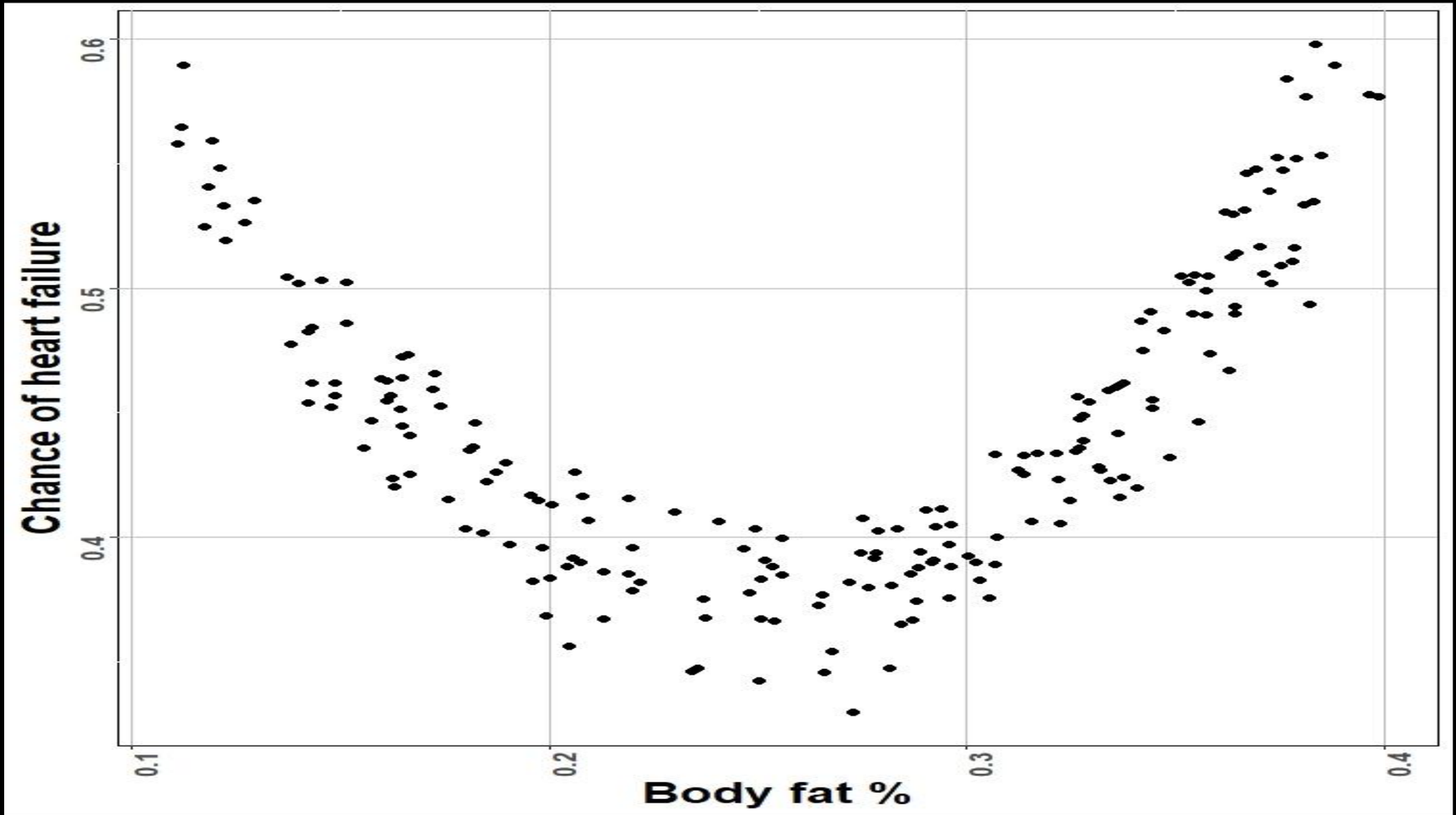
Displaying the data

When both the DV and IV are numerical, we can represent data in the form of a scatterplot.



Displaying the data

It is important to perform a scatterplot because it helps us to see if the relationship is linear.



In this example, the relationship between body fat % and chance of heart failure is not linear and hence it is not sensible to use linear regression.

Linear model

In regression, the relationship between Y and X is modelled in the following form:

$$\mathbf{Y = a + b * X + E}$$

where:

- **Y** is the dependent variable (Income in the example)
- **X** is the independent variable (IQ in the example)
- **a** is an intercept
- **b** is the coefficient
- **E** is an error term for each observation (since there is additional variation not explained by income)

Assumptions of regression

- The errors E are normally distributed.

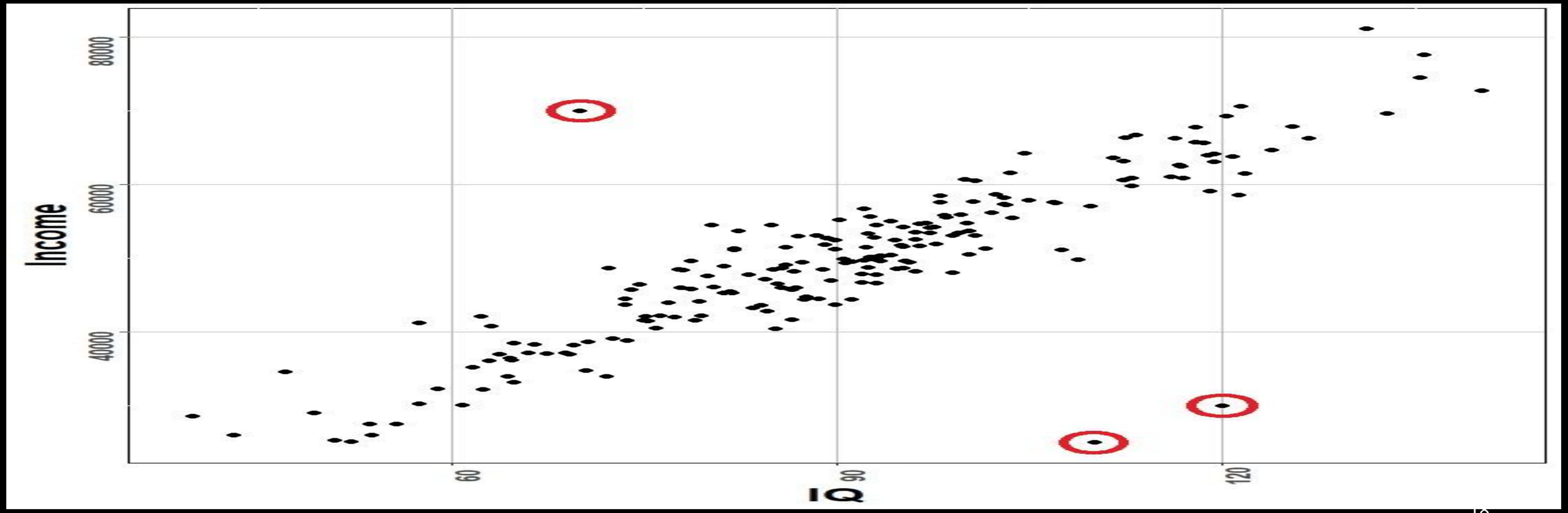
This can be tested by plotting an histogram of the residuals of the regression and checking that they all have a bell shape.

Alternatively, you could use the Shapiro-Wilk test for normality.

Assumptions of regression

- There are no clear outliers

This can be checked by performing the scatterplot. The outliers (circled in red in the figure) can simply be removed from the analysis .



Linear model

We are not interested in the intercept **a** but only in the coefficient **b**.

The coefficient **b** represents the relationship between X and Y.

- If b is positive, X has a positive effect on Y (as X increases, Y increases);
- If b is negative, X has a negative effect on Y (as X increases, Y decreases).

If b = 0, there is **no effect** of X on Y.

Terminologies Related to the Regression Analysis:

- **Dependent Variable**
- **:** The main factor in Regression analysis which we want to predict or understand is called the dependent variable. It is also called **target variable**.

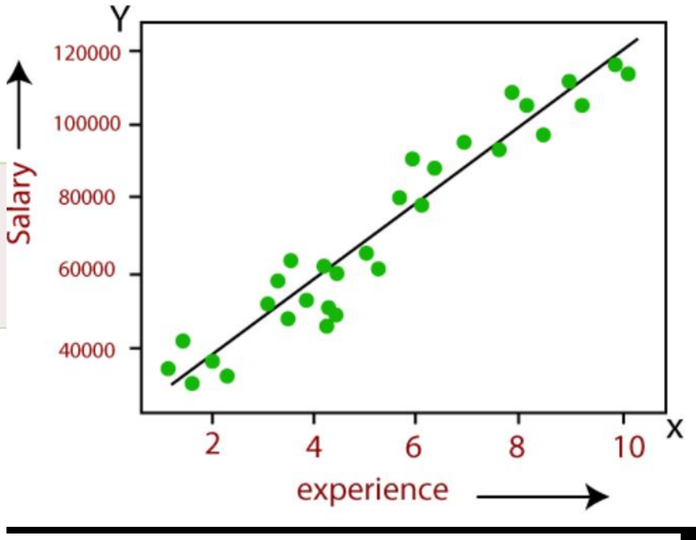
- **Independent Variable**
- **:** The factors which affect the dependent variables or which are used to predict the values of the dependent variables are called independent variable, also called as a **predictor**.

- **Outliers:** Outlier is an observation which contains either very low value or very high value in comparison to other observed values
- . An outlier may hamper the result, so it should be avoided.

- **Multicollinearity:** If the independent variables are highly correlated with each other than other variables, then such condition is called Multicollinearity. It should not be present in the dataset, because it creates problem while ranking the most affecting variable.

- **Underfitting and Overfitting**
- **:** If our algorithm works well with the training dataset but not well with test dataset, then such problem is called **Overfitting**. And if our algorithm does not perform well even with training dataset, then such problem is called **underfitting**.

- Below is the mathematical equation for Linear regression:



$$Y = aX + b$$

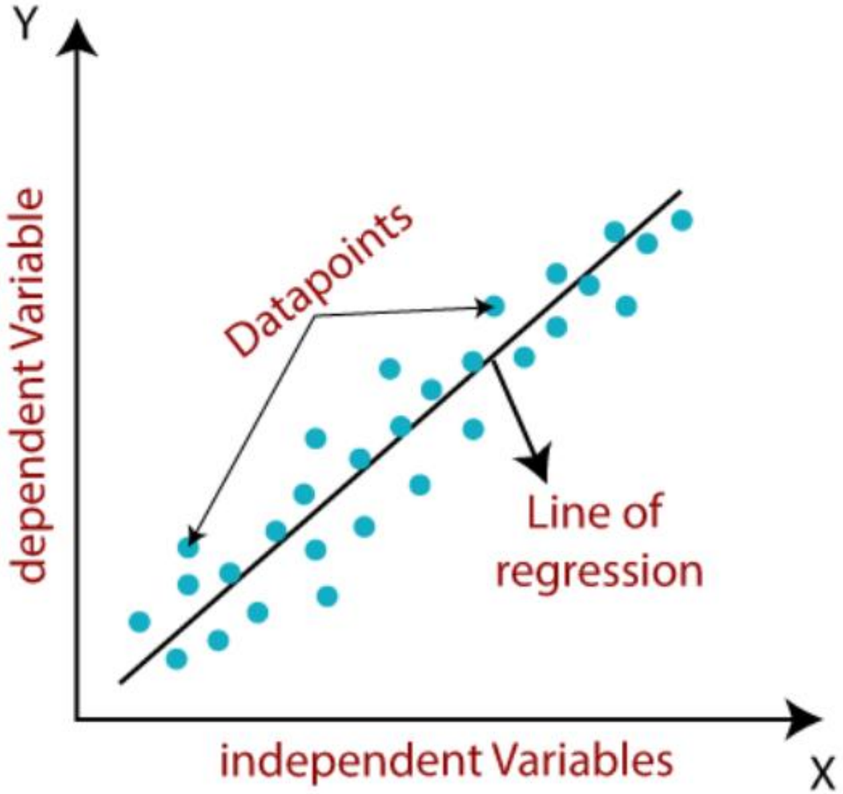
Here, Y = dependent variables (target variables),

X= Independent variables (predictor variables),

a and b are the linear coefficients

Some popular applications of linear regression are:

- **Analyzing trends and sales estimates**
- **Salary forecasting**
- **Real estate prediction**
- **Arriving at ETAs in traffic.**



Types of Linear Regression

- Linear regression can be further divided into two types of the algorithm:
- **Simple Linear Regression:**

  If a single independent variable is used to predict the value of a numerical dependent variable, then such a Linear Regression algorithm is called Simple Linear Regression.
- **Multiple Linear regression:**

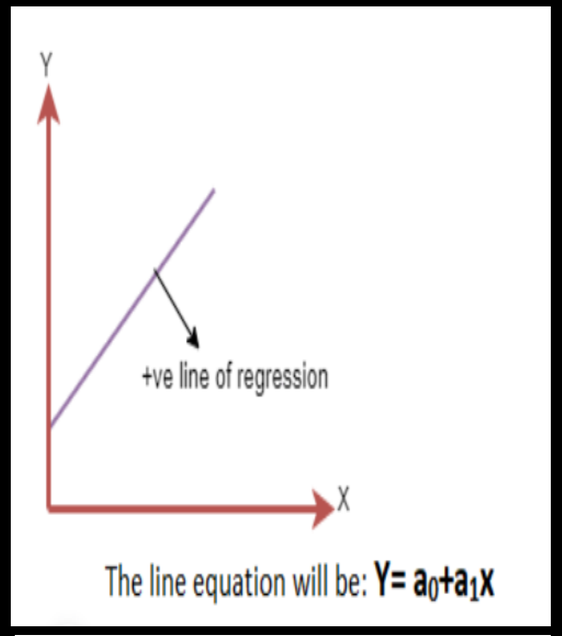
  If more than one independent variable is used to predict the value of a numerical dependent variable, then such a Linear Regression algorithm is called Multiple Linear Regression.

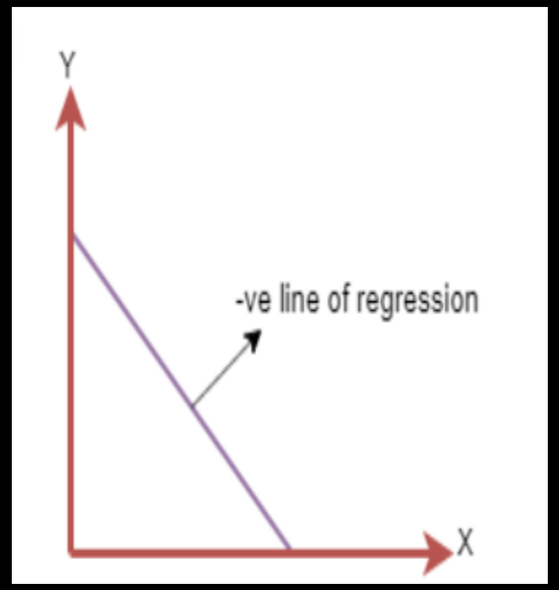
Linear Regression Line

- A linear line showing the relationship between the dependent and independent variables is called a **regression line**. A regression line can show two types of relationship:

- **Positive Linear Relationship:**
  If the dependent variable increases on the Y-axis and independent variable increases on X-axis, then such a relationship is termed as a Positive linear relationship.

- **Negative Linear Relationship:**
  If the dependent variable decreases on the Y-axis and independent variable increases on the X-axis, then such a relationship is called a negative linear relationship.





The Model

The model has a deterministic and a probabilistic components

House
Cost

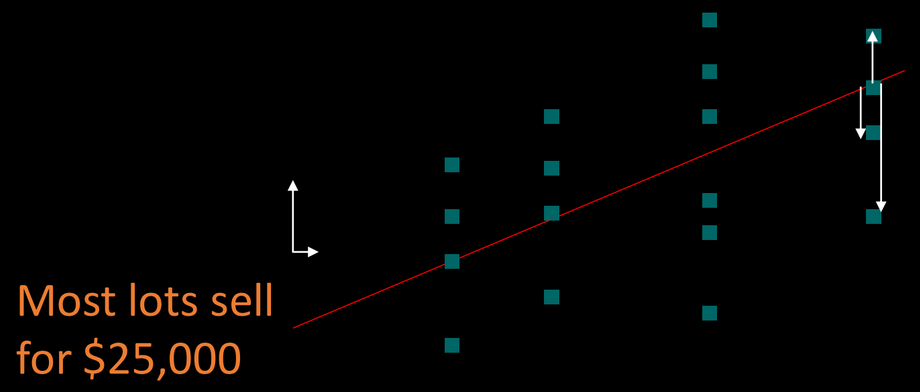
Building a house costs about
$75 per square foot.
House cost = 25000 + 75(Size)

Most lots sell
for $25,000

House size

However, house cost vary even among same size houses!

Since cost behave unpredictably, we add a random component.



House cost = 25000 + 75 (Size) + ε

House size

- The first order linear model

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

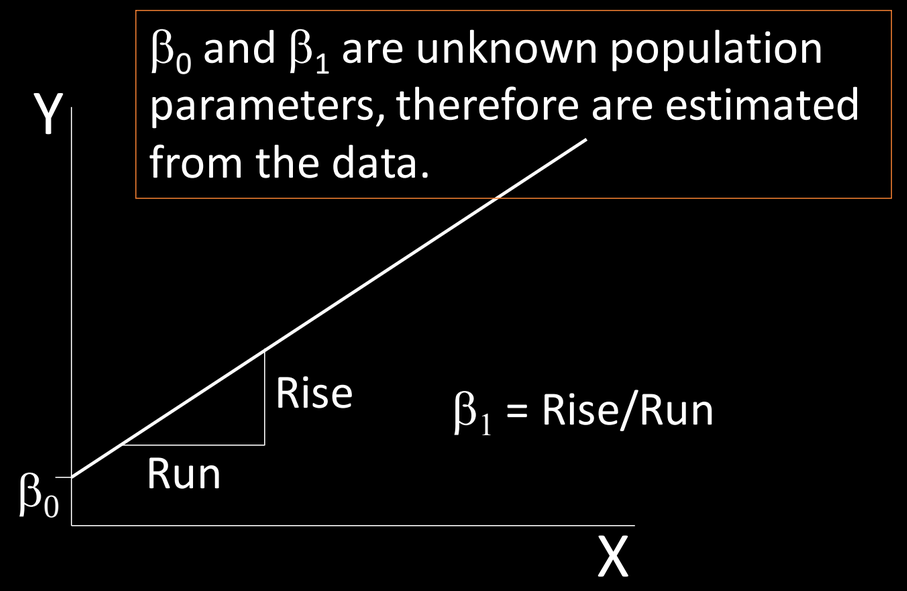
Y = dependent variable

X = independent variable

β_0 = Y-intercept

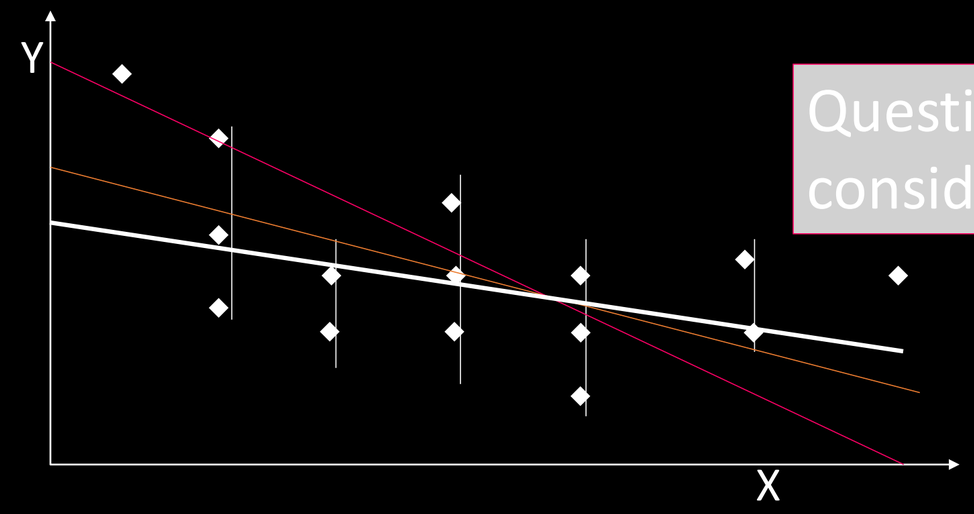
β_1 = slope of the line

ε = error variable



Estimating the Coefficients

- The estimates are determined by
  - drawing a sample from the population of interest,
  - calculating sample statistics.
  - producing a straight line that cuts into the data.



Question: What should be considered a good line?

The Least Squares (Regression) Line

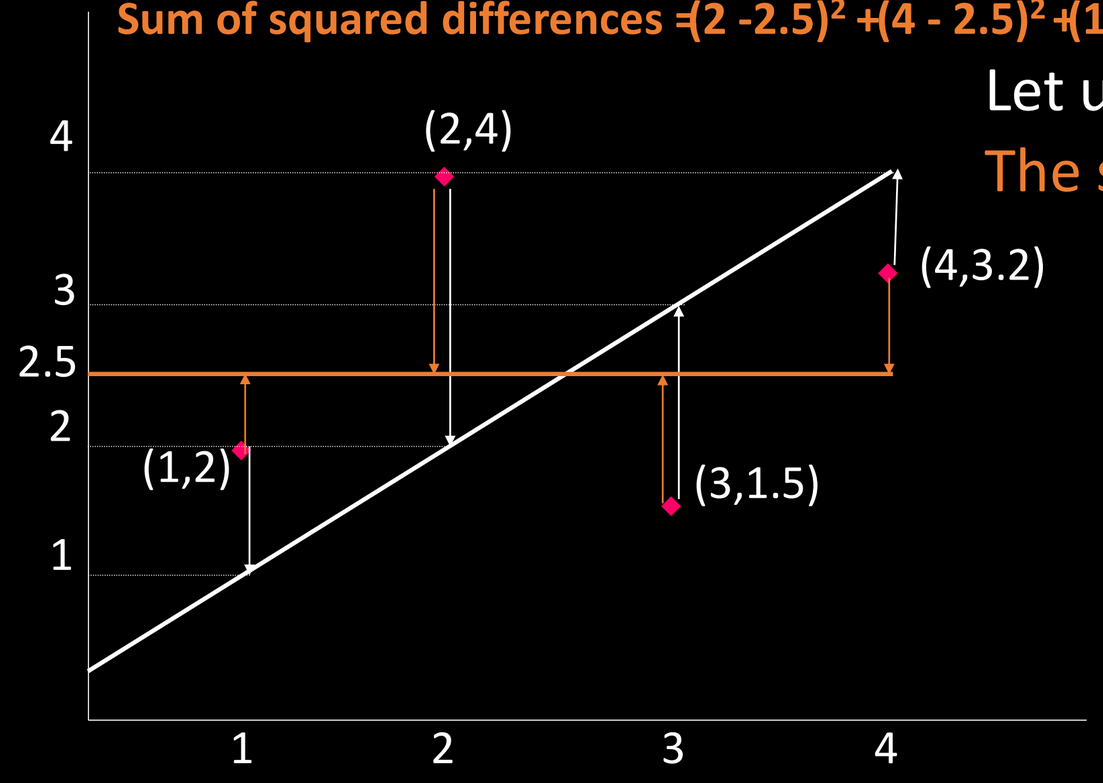
A good line is one that minimizes
the sum of squared differences between the points and the line.

Sum of squared differences =(2 - 1)^2 +(4 - 2)^2 +(1.5 - 3)^2 +(3.2 - 4)^2 = 6.89

Sum of squared differences =(2 -2.5)^2 +(4 - 2.5)^2 +(1.5 - 2.5)^2 +(3.2 - 2.5)^2 = 3.99

Let us compare two lines

The second line is horizontal



The smaller the sum of squared differences the better the fit of the line to the data.

The Estimated Coefficients

To calculate the estimates of the line coefficients, that minimize the differences between the data points and the line, use the formulas:

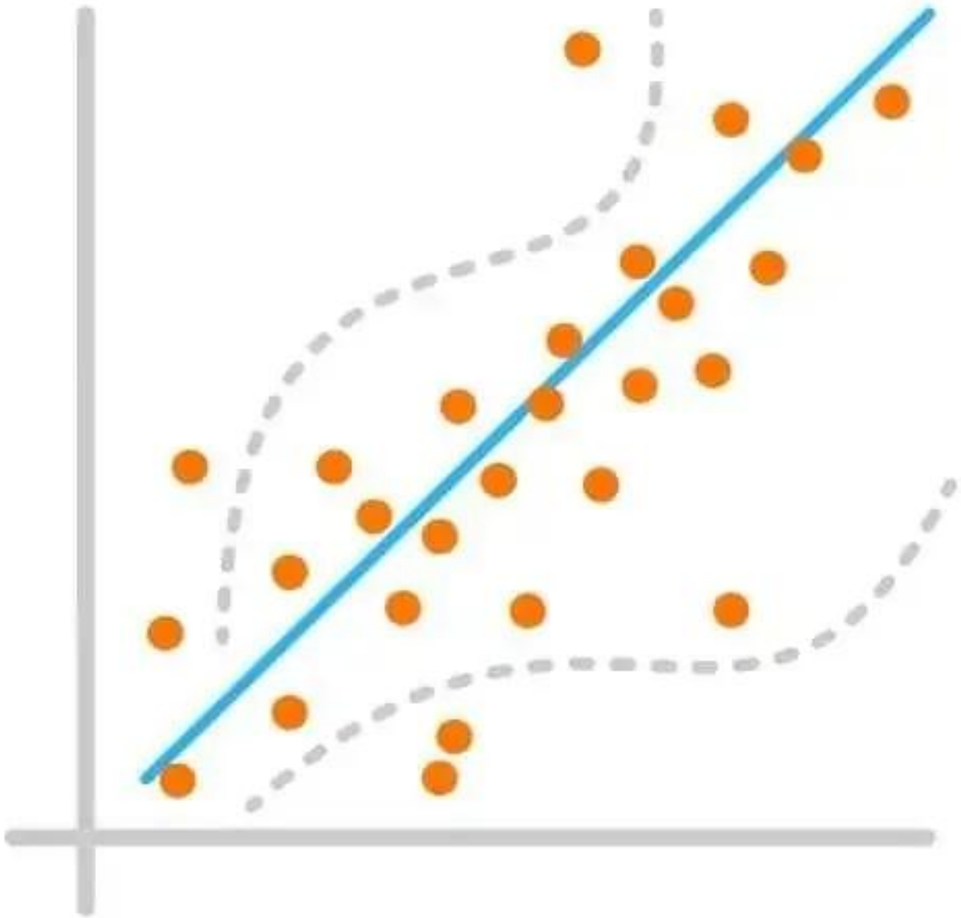
$$b_1 = \frac{\mathrm{cov}(X,Y)}{s_X^2}\left(= \frac{s_{XY}}{s_X^2}\right)$$

$$b_0 = \overline{Y} - b_1\overline{X}$$

The regression equation that estimates the equation of the first order linear model is:

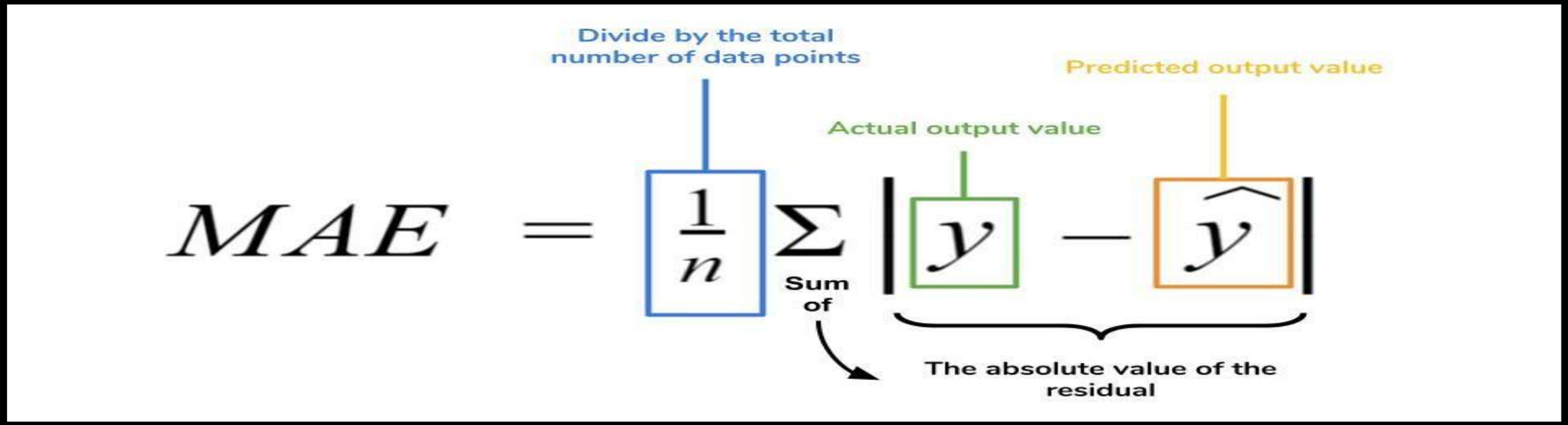
$$\hat{Y} = b_0 + b_1 X$$

4 Common Regression Metrics



1. Mean Squared Error (MSE)
2. Root Mean Squared Error (RMSE)
3. Mean Absolute Error (MAE)
4. R-squared (R^2)

Evaluation Metrics: Mean Absolute Error



Meaning: Average of absolute differences between actual and predicted values.
Interpretation: Average error in original units.

Pros: Easy to understand, robust to outliers compared to squared errors
Cons: Does not penalize large errors strongly

Evaluation Metrics: Mean Square Error

Mean

Error

Squared

$$\mathrm{MSE} = \boxed{\frac{1}{n}\sum_{i=1}^{n}}\left(\boxed{Y_i - \hat{Y}_i}\right)^{\boxed{2}}$$

Meaning: Average of squared differences between actual and predicted values.

Pros: Penalizes large errors heavily

Cons: Units are squared, harder to interpret

Evaluation Metrics: Root Mean Square Error

$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{(\hat{y}_i - y_i)^2}{n}}$$

Meaning: Square root of MSE, brings error back to original units.
Pros: Interpretable like MAE, but still penalizes large errors
Cons: Sensitive to outliers

Evaluation Metrics: Mean Absolute Percentage Error

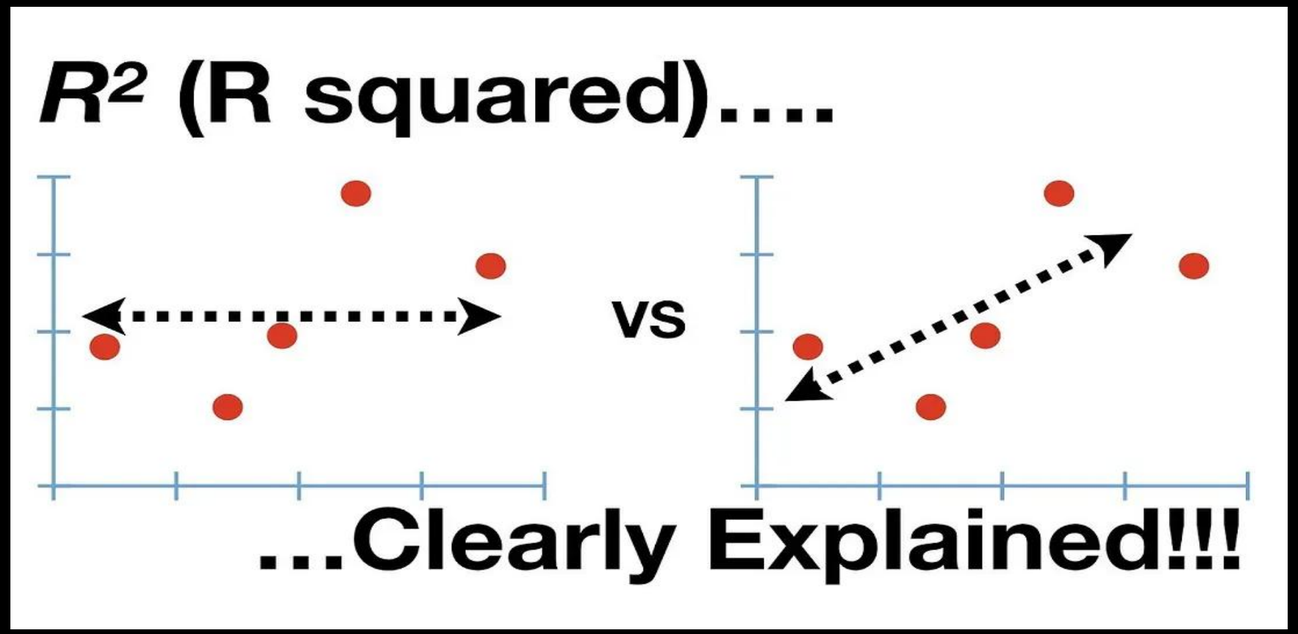
Multiplying by 100% converts to percentage

The residual

$$MAPE = \frac{\boxed{100\%}}{n} \sum \left| \frac{\overbrace{y - \hat{y}}^{\text{The residual}}}{\boxed{y}} \right|$$

Each residual is scaled against the actual value

Evaluation Metrics: R-squared



$$1 - r^2 = 1 - \frac{\text{RSS}}{\text{TSS}}$$

$$= 1 - \frac{\sum(\hat{Y} - \bar{Y})^2}{\sum(Y - \bar{Y})^2}$$

Meaning: Measures how well the model explains variance in data.

Range:

- 1 → perfect model
- 0 → model is as good as mean prediction
- < 0 → worse than mean model

Pros: Good for overall performance comparison

Cons: Can be misleading for non-linear or poorly scaled data5.

Adjusted R^2:

- Modified version of R^2 that accounts for number of features
- Penalizes unnecessary predictors

THANK YOU